GG-B-823C
June 17, 1975
SUPERSEDING
Fed. Spec. GG-B-823B
August 12, 1957 and
MIL-B-36179A
September 9, 1965

FEDERAL SPECIFICATION

BURS, DENTURE TRIMMING, HIGH SPEED TOOL STEEL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers denture trimming burs used for finishing dentures in prosthetic laboratory procedures.

1.2 Classification.

1.2.1 Shapes. Denture trimming burs shall be of the following shapes, as specified (see 6.1):

Shape A. Flame, No. 1
Shape B. Pear, No. 3
Shape C. Round, No. 2

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids of request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

FSC 6520

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Federal Standards:

FED-STD-151	Metal, Test Methods.
FED-STD-123	Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116	Methods of Preservation.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual, Tool Steels

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Material. The burs shall be made of AISI type M2 high speed tool steel.

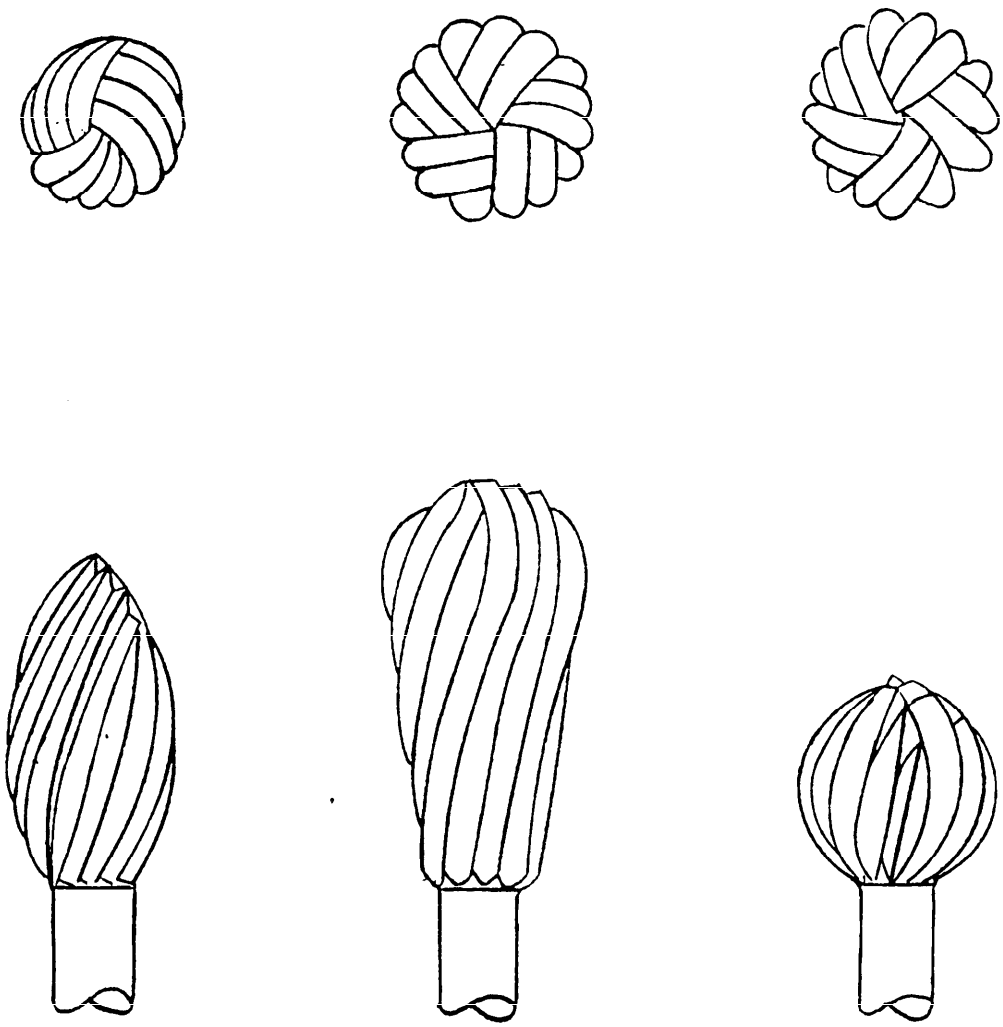
3.2 Bur dimensions. The bur shall conform to Table I for the dimensions specified for the applicable shape.

Table I. Bur Dimensions (inch)

	SHAPES		
	A. Flame	B. Pear	C. Round
Overall length, ± 1/8	1-3/4	1-3/4	1-5/8
Shank diameter, ± 0.0005	0.0924	0.0924	0.0924
Number of blades, ± 1	12	15	15
Head diameter at largest diameter, ± 0.005	0.188	0.245	0.245
Length of head, ± 1/64	27/64	13/32	1/4

3.3 Design and construction. The bur shape shall be as shown in Figure 1. The burs shall be denture trimming burs, suitable for use with straight handpieces in finishing plastic denture materials in prosthetic laboratory procedures. Burs shall be furnished in shapes as specified. The burs shall be of one-piece construction, and shall consist of a shank and head. The shank of bur shall be straight. The blades of the bur head shall be shaped and sharpened to cut at the tip as well as the sides of the bur head, and provide automatic chip clearance.

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SHAPE OF BUR HEADS

FIGURE 1

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3.4 Runout. The total indicated runout of the burs shall not exceed 0.003 inch, when tested as specified in 4.4.2.

3.5 Hardness. The head and shank of the burs shall have a minimum Rockwell hardness of C62, when tested as specified in 4.4.3.

3.6 Fatigue in cutting. The time required to cut the eighteenth half-inch shall be not more than 2.25 the time that is required to cut the second half-inch, when tested as specified in 4.4.4.1 and 4.4.4.2. The burs shall show no bending as a result of these tests.

3.7 Finish. The burs shall have a black oxidized finish. The shank of the burs shall be smoothly finished.

3.8 Marking. Marking of the burs with the name or trademark shall be at the option of the supplier.

3.9 Workmanship. Workmanship shall be first class throughout. The burs shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, IV and V (see 4.2.1, 4.3 and 4.3.3) and table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. Unit of product for examination purposes shall be one bur.

Table II. Sampling for Examination

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified in Table III. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one bur.

Table III. Test criteria

Characteristic	Requirement	Test procedure	Inspection level
Chemical composition	3.1	4.4.1	(see 4.2.3)
Runout	3.4	4.4.2	S-3
Hardness	3.5	4.4.3	S-2
Fatigue in cutting (side of head)	3.6	4.4.4.1	S-2
Fatigue in cutting (tip of head)	3.6	4.4.4.2	S-2

4.2.3 Raw material. A sufficient quantity of each lot of raw material used in the manufacture of the burs shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same burs shall be used for the determination of two or more test characteristics.

4.3 Examination. The burs shall be examined for the possible defects listed in Table IV. The examination shall not be limited to those defects shown in the table. Unless otherwise indicated in the table, the defect listed shall apply to all shapes of tungsten steel, denture trimming burs covered by this specification.

Table IV. Classification of defects

Categories	Defects
Major A:	
101	Bur shape not as specified.
102	Bur shank not as specified.
103	Diameter of shank not as specified.
104	Bur blades not cleanly cut.

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Table IV. Classification of Defects (cont'd)

Categories	Defects
Major B:	
151	Head diameter not as specified.
152	Length of head not as specified.
Minor A:	
201	Number of blades not as specified.
202	Finish on bur not as specified.
203	Packaging, packing, and marking not as specified.
Minor B:	
251	Overall length of bur not as specified.

4.3.1 Dimensions. Examination of the dimensions, as listed in 3.2, shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

4.3.2 Microscopic examination. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 10 power (10X) magnification.

4.3.3 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table V. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging & packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

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4.4 Tests.

4.4.1 Chemical composition. In the absence of certificates of quality for the raw material, the samples, selected as specified in 4.2.3, shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis.

4.4.2 Runout. The bur shall be placed and held in a "V" block or prescision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.4.3 Hardness. Hardness shall be determined in accordance with Method 243 (Rockwell) or Method 244 (Diamond Pyramid) of FED-STD-151.

4.4.4 Fatigue in cutting.

4.4.4.1 Side of head. Tests shall be made on a milling machine or similar device. The feed shall be that produced by a 1-pound weight operating a cord attached to a carriage mounted on ball bearings. The bur shall rotate at 1,600 revolutions per minute. The material to be cut shall be 0.020 rolled brass with a Brinell hardness of 74-81 (Rockwell 30-T, 42-49). The edge of the brass shall be centered on the length of the head of the bur. The bur shall be perpendicular to the brass sheet, and the cut shall be made the length of the brass. A recording device operated by a synchronous motor shall be used to time the cut.

4.4.4.2 Tip of head. Test shall be conducted as in 4.4.4.1, except that brass test piece shall be applied against the point of the bur.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing six burs of the shape specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

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5.1.2.2 Preservative. Preservative shall not be utilized.

5.1.2.3 Unit package. Six burs of one shape and size, as specified, shall be packaged in a box of appropriate size so designed that each bur is contained in an individual cell. The box shall be constructed of an anticorrosive material that will not interact with the contents. Closure of the box shall be secured to prevent spilling of contents under normal handling.

5.1.2.4 Intermediate package. Twelve unit packages shall be packed in snug-fitting box constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. Closure of the box shall be secured to prevent spilling of contents under normal handling.

5.1.3 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Ninety-six intermediate packages (1152 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Special requirements as specified in PPP-B-636 shall apply for fiberboard boxes.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.2 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.3 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. Ninety-six intermediate packages (1152 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type;

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PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636. Special requirements as specified in PPP-B-636 shall apply for fiberboard boxes.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Shape required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

Stock No.	Item Identification
6520-00-076-8682	BUR, DENTURE TRIMMING, Straight Handpiece, High Speed Tool Steel, 6s: Flame Shape, No. 1
6520-00-076-8683	BUR, DENTURE TRIMMING, Straight Handpiece, High Speed Tool Steel, 6s: Round Shape, No. 2
6520-00-076-8684	BUR, DENTURE TRIMMING, Straight Handpiece, High Speed Tool Steel, 6s: Pear Shape, No. 3

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1378

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

GG-B-823C - Burs, Denture Trimming, High Speed Tool Steel

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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